

NexRay AI

Medical Assistant Platform • Clinical Decision Support Report

25 July 2026 | 04:57 PM

Session ID: 30

Modules Used: Symptom Checker •
Refined Diagnosis

Date: 25 July 2026 | 04:57 PM

Symptom Analysis

Possible Condition	Confidence
Bacterial Meningitis	92%
Viral Meningitis	78%
Cerebral Malaria	65%

■ **Urgency Level: Emergency — Immediate action required**

Recommended Tests

- Lumbar puncture with cerebrospinal fluid (CSF) analysis including cell count, protein, glucose, and Gram stain
- Blood culture to identify causative organism
- Rapid malaria test (RDT) or blood smear microscopy
- Complete blood count (CBC)
- Blood glucose level
- Rapid plasma reagin (RPR) if syphilis suspected

Suggested Treatment

- Empiric broad-spectrum antibiotics (ceftriaxone 2g IV every 12 hours plus vancomycin) pending culture results if bacterial meningitis suspected
- Supportive care including fluid management, fever control, and seizure prophylaxis if needed
- Antimalarial therapy if malaria is confirmed or suspected pending meningitis exclusion

Next Steps

- Immediate admission and isolation precautions
- Perform lumbar puncture urgently to differentiate bacterial vs viral meningitis
- Initiate empiric antibiotic therapy while awaiting CSF results
- Administer antimalarial therapy given West African context if malaria cannot be excluded

Summary: The patient presents with a classic meningitis syndrome with fever, severe headache, neck stiffness, photophobia, and vomiting. Bacterial meningitis is the most likely diagnosis and constitutes a medical emergency. Viral meningitis and cerebral malaria must be ruled out. Immediate lumbar puncture and empiric antibiotic therapy are critical. Given the West African context, concurrent malaria testing is essential.

Refined Diagnosis — Based on Test Results

Final Diagnosis: Bacterial meningitis caused by Neisseria meningitidis (meningococcal meningitis), confirmed by characteristic CSF findings and Gram stain identification of gram-negative diplococci.

■ Confirmed Conditions

• **Bacterial Meningitis** (Confirmed) — Cloudy CSF under high pressure, markedly elevated WBC (2800) with 95% neutrophils, elevated protein (3.2g/L), and critically low glucose (1.1mmol/L with CSF:blood ratio of 0.19). Gram-negative diplococci on Gram stain is highly suggestive of Neisseria meningitidis. Elevated blood WBC (18,500) with neutrophilia supports bacterial infection. This constellation of findings is pathognomonic for bacterial meningitis.

■ Ruled Out

- **Viral Meningitis** — CSF profile is incompatible with viral meningitis. While viral meningitis can present with similar symptoms, the markedly elevated neutrophilic response (95% neutrophils), very low CSF glucose, and high protein levels are characteristic of bacterial rather than viral infection.
- **Cerebral Malaria** — Malaria rapid test is negative, ruling out active malaria parasitemia. The CSF findings (neutrophilic pleocytosis, low glucose, elevated protein) are not consistent with cerebral malaria pathophysiology.

■ **Urgency Level: Emergency — Immediate action required**

Updated Treatment

- Immediate initiation of empiric antibiotic therapy: Ceftriaxone 2g IV every 12 hours (or cefotaxime if available) PLUS Vancomycin 15-20mg/kg IV every 8-12 hours for CNS penetration
- Dexamethasone 10mg IV every 6 hours for 4 days (if not already given, administer before or with first antibiotic dose to reduce mortality and neurological sequelae)
- Supportive care: IV fluid management with careful monitoring of electrolytes (risk of SIADH), maintain adequate oxygenation and airway protection
- Contact prophylaxis: Identify and treat close contacts with rifampicin or ceftriaxone as per local guidelines
- Anticipate blood culture results to guide definitive antibiotic selection and duration

Next Steps

- Monitor blood culture results when available to confirm organism and perform sensitivity testing for antibiotic optimization
- Perform repeat lumbar puncture at 24-48 hours if clinical deterioration or inadequate response to assess treatment efficacy
- Monitor for complications: increased intracranial pressure, subdural effusion, ventriculitis, seizures, and hearing loss
- Arrange audiological assessment post-recovery (within 4 weeks) as sensorineural hearing loss is a significant sequela
- Institute meningococcal vaccination if not previously given (for both patient post-recovery and close contacts)

Summary: The test results definitively confirm bacterial meningitis, most likely caused by *Neisseria meningitidis*. The CSF profile is pathognomonic with neutrophilic pleocytosis, markedly elevated protein, severely depressed glucose, and direct visualization of gram-negative diplococci. Malaria has been excluded. This is a medical emergency requiring immediate high-dose bactericidal antibiotics with CNS penetration and adjunctive dexamethasone therapy. Prognosis depends on rapidity of treatment initiation and supportive care quality.

■ **DISCLAIMER:** This report is generated by NexRay AI and is intended solely as a clinical decision-support tool. All findings, conditions, treatments, and recommendations are AI-generated suggestions and do not constitute a formal medical diagnosis or prescription. The clinical judgment of the attending medical professional must be applied before any action is taken.